

# APPLIED STATISTICS, PROBABILITY THEORY AND MATHEMATICAL STATISTICS

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Linear regression

# Linear regression

Let the linear regression model of  $Y$  and  $X$  be

$$y = ax + b.$$

If the observed values of  $(x, y)$  are  $(x_i, y_i)$ , then the error values are given by

$$\epsilon_i = y_i - ax_i - b, \quad i = 1, \dots, n.$$

Using the least square estimation we have to minimize the following function:

$$\sum_{i=1}^n \epsilon^2 = \sum_{i=1}^n (y_i - ax_i - b)^2$$

Using the denotions:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i, \quad \mathbf{s}_1^{*2} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad \mathbf{s}_2^{*2} = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2, \quad \mathbf{m}_{11}^* = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

The estimations for  $a$  and  $b$  are:

$$\hat{a} = \frac{\mathbf{m}_{11}^*}{\mathbf{s}_1^{*2}}, \quad \hat{b} = \bar{y} - \bar{x} \frac{\mathbf{m}_{11}^*}{\mathbf{s}_1^{*2}}.$$

# Accidents

```
load accidents
```

```
x = hwydata(:,14); % Population of states
```

```
y = hwydata(:,4); % Accidents per state
```

```
a = corr(x,y) / std(x) * std(y);
```

```
b = mean(y) - mean(x)*a;
```

```
scatter(x,y)
```

```
hold on
```

```
plot(x, x*a + b)
```

```
xlabel('Population of state')
```

```
ylabel('Fatal traffic accidents per state')
```

```
title('Linear regression relation between accidents & population')
```

```
grid on
```

# Carsmall

**load carsmall**

**X = [Weight, Horsepower, Acceleration];**

**mdl = fitlm(X,MPG)**

**mdl.Coefficients**

# Count

```
load count.dat
```

```
x = count(:,1);
```

```
y = count(:,2);
```

```
pl = polyfit(x,y,1) % Linear
```

```
pq = polyfit(x,y,2) % Quadratic
```

```
scatter(x,y)
```

```
hold on
```

```
z = linspace(min(x),max(x),100);
```

```
plot(z, polyval(pl,z))
```

```
plot(z, polyval(pq,z))
```

# height-weight

```
%regr height-weight
n=10;
x=[160 166 170 175 178 179 180 185 190 198];
y=[ 55  70  65  80  75  85  80  92  85  97];
%approximate y by a linear function of x
%using the theorem
mx=mean(x);
my=mean(y);
vx=mean(x.^2)-mx^2;
vy=mean(y.^2)-my^2;
mxy=mean(x.*y)-mx*my;   %cov
a=mxy/vx
b=my-mx*a
figure
plot(x,y,'r*');
hold on
t=150:0.01:210;
plot(t,a*t+b,'b-');
hold on
%using the Matlab program fitlm
X = [x];   %linear
mdl1 = fitlm(X,y)
%mdl1.Coefficients
%approximate y by a quadratic function of x
X = [x',x'.^2];   %quadratic
mdl2 = fitlm(X,y')
a=mdl2.Coefficients.Estimate
plot(t,a(1)+a(2)*t+a(3)*t.^2,'m-');
```

# accidents

```
load accidents
x = hwydata(:,14); % Population of states
y = hwydata(:,4); % Accidents per state
%approximate y by a linear function of x
%using the theorem
a = corr(x,y) / std(x) * std(y);
b = mean(y) - mean(x)*a;
scatter(x,y)
hold on
plot(x, x*a + b)
xlabel('Population of state')
ylabel('Fatal traffic accidents per state')
title('Linear regression relation between accidents & population')
grid on
%using the Matlab program fitlm
X = [x]; %linear
mdl1 = fitlm(X,y)
%mdl1.Coefficients
a0=mdl1.Coefficients.Estimate
t=0:1:3.5*10^7;
figure
scatter(x,y);
hold on
plot(t,a0(1)+a0(2)*t,'r-');
hold on
%approximate y by a quadratic function of x
X = [x';x'.^2]; %quadratic
mdl2 = fitlm(X',y)
%mdl2.Coefficients
a1=mdl2.Coefficients.Estimate
plot(t,a1(1)+a1(2)*t+a1(3)*t.^2,'b-');
%approximate y by a quadratic function of x
mdl3 = fitlm(x,y,'quadratic'); %quadratic2
a2=mdl3.Coefficients.Estimate
%approximate y by a quadratic function of x using robust loss function
mdl4 = fitlm(x,y,'quadratic','RobustOpts','huber'); %quadratic3
a3=mdl4.Coefficients.Estimate
```

# count

```
load count.dat
x = count(:,1);
y = count(:,2);
%approximate y by a linear function of x using polyfit
pl = polyfit(x,y,1) % Linear
%approximate y by a quadratic function of x using polyfit
pq = polyfit(x,y,2) % Quadratic
figure
scatter(x,y)
hold on
z = linspace(min(x),max(x),100);
plot(z, polyval(pl,z))
plot(z, polyval(pq,z))
figure
%approximate y by a linear and a quadratic function of x using fitlm
scatter(x,y)
hold on
mdl = fitlm(x,y)
a=mdl.Coefficients.Estimate
plot(z,a(1)+a(2)*z,'r-');
mdl1 = fitlm(x,y,'quadratic')
a1=mdl1.Coefficients.Estimate
plot(z,a1(1)+a1(2)*z+a1(3)*z.^2,'b-');
```

# Literature

- ❑ Ágnes Baran: Mathematics for engineers 1. (Laboratory slides)
- ❑ Sándor Baran: Probability theory and statistics
- ❑ Fazekas István: Valószínűségszámítás, Debreceni Egyetemi Kiadó, 2009
- ❑ Fazekas István: Bevezetés a matematikai statisztikába, Debreceni Egyetemi Kiadó, 2009
- ❑ Matlab examples:  
<https://www.mathworks.com/help/examples.html>